

— MULTI-AGENT DUE-DILIGENCE FOR COMMERCIAL REAL ESTATE

The institutional diligence committee, run by **agents**.

*eight distinct agents —
not one model in costumes*

Eight specialist agents across two frameworks share one Band room — reading each other's findings, resolving contradictions, delegating tasks, and recruiting quantitative models on demand to produce a defensible deal memo.

— THE PROBLEM

Real-estate diligence is a committee of experts — and the committee is the bottleneck.

Acquiring a single asset requires title, zoning, environmental, structural, insurance and financial review. Today that means weeks of email, siloed PDFs, and an associate manually reconciling expert findings before capital is committed.

SLOW & SERIAL

3–6 wks

to clear diligence on a mid-market asset, most of it lost to hand-offs between specialists.

COSTLY TO GET WRONG

\$100M+

a single missed easement, zoning conflict or remediation liability can move an IRR below water.

NO AUDIT TRAIL

Opaque

the rationale for an approval lives across inboxes — difficult to replay or defend to an IC.

Our thesis

The committee can run in minutes if the experts share one context, surface contradictions between themselves, and leave a complete, replayable record of every finding, delegation and number.

— THE SOLUTION

A standing committee of agents that coordinate through Band.

- 01 **Read the room.** Each agent pulls the live Band conversation as its primary context before it reasons.
- 02 **Challenge.** When two agents disagree — an easement one finds and another misses — they resolve it in-thread.
- 03 **Delegate.** Any agent can assign another a defined task, with intent and authority, derived from context.
- 04 **Recruit.** The committee brings Python quantitative specialists into the room when the deal warrants it.
- 05 **Synthesize.** A Deal Director composes a Red / Yellow / Green memo with conditions precedent.

OUTCOME

A defensible deal memo with a complete audit trail — in minutes, not weeks.

DECIDED AT RUNTIME

Which agents debate, who delegates to whom, and which specialists are recruited are all **determined from the deal context** — not a fixed script. The collaboration is emergent.

Eight agents. Two frameworks. One room.

● TypeScript · reasoning

● Python · quantitative

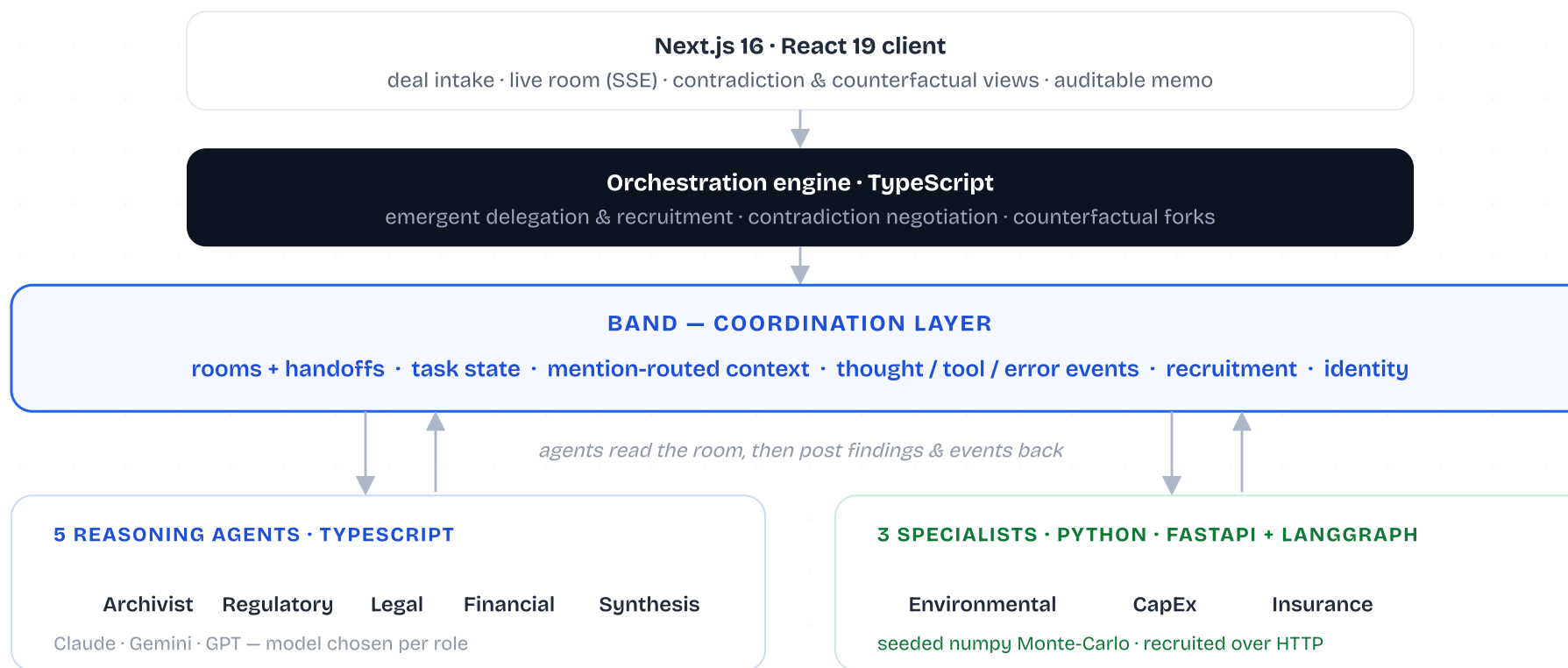
● Archivist Document intake → structured property facts and missing-document gaps.	● Regulatory Zoning, permits, FEMA flood and deed restrictions, severity-ranked.	● Legal Risk Title defects, easement conflicts, liens and seller-representation gaps.	● Financial NOI, cap rate, DSCR and a 5-year IRR — computed deterministically.
● Synthesis Deal Director — composes the R / Y / G memo and conditions precedent.	● Environmental Contamination risk and Monte-Carlo remediation cost; Phase I call.	● CapEx / Construction Renovation and conversion cost with schedule risk, simulated.	● Insurance / Catastrophe Flood, wind and seismic exposure with a premium estimate.

Why two frameworks?

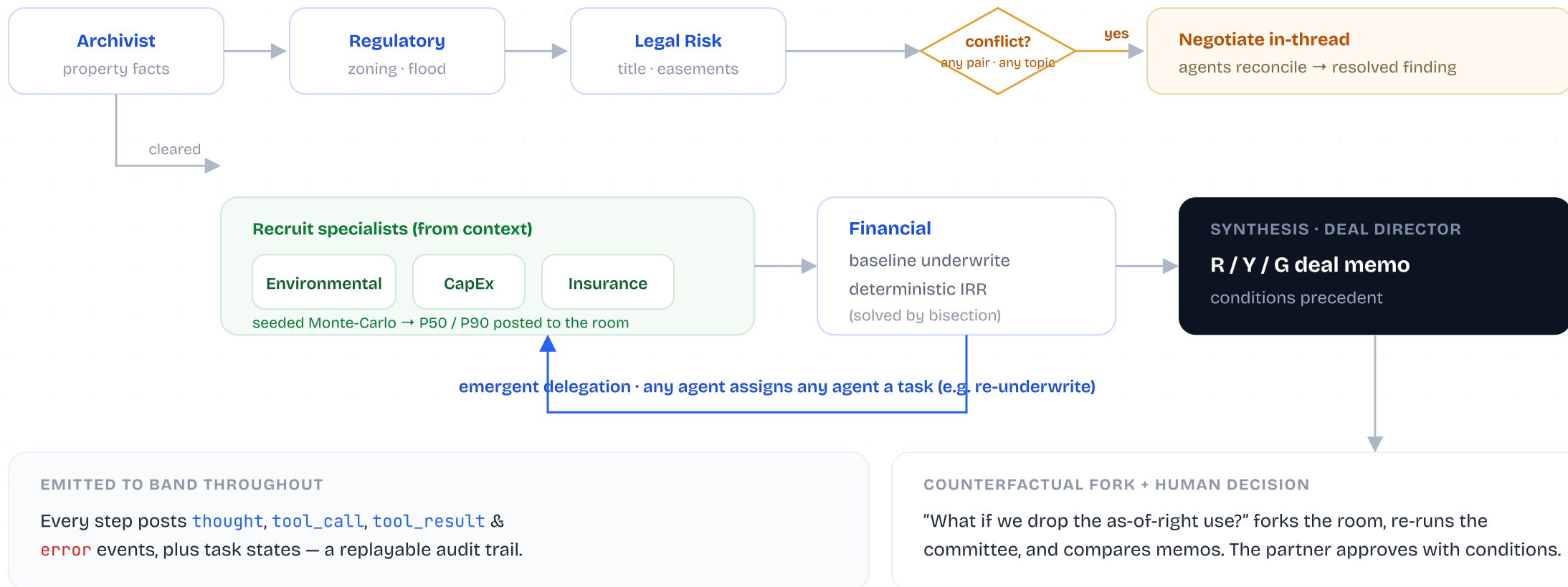
The reasoning agents live in TypeScript alongside the product. The specialists are Python because they perform genuine **numerical work** — seeded numpy Monte-Carlo over LangGraph state machines. Band unifies them into one committee.

Band is the spine — not a wrapper.

The orchestrator drives the workflow, but every agent communicates *through* Band: reading shared context, posting findings, and emitting events.



One deal, end to end.



Every Band primitive is load-bearing.

Remove Band and the committee stops being a committee. These are the primitives the system depends on.

Shared context Agents read the room via mention-routed getContext before reasoning — primary context, not spoon-fed payloads.	Handoffs Structured, mention-based posting passes the baton along the committee, each agent addressing the next.	Task state Delegations are real Band tasks tracked processing → processed — accountable, not fire-and-forget.
Rich events thought · tool_call · tool_result · error make each agent's reasoning visible and replayable.	Recruitment Specialists are pulled into the live room mid-deal — cross-framework agents joining as first-class participants.	Distinct identity Each specialist posts under its own Band identity (per-agent key) — genuinely separate agents.

Collaboration is data, not hardcoded branches.

lib/agents/schemas.ts

```
// each agent emits these from the deal context
export const ComplianceReportSchema = z.object({
  // ...findings, risk_score...
  delegations: z.array(DelegationRequestSchema),
  requested_specialists: z.array(SpecialistRequestSchema),
});

const DelegationRequestSchema = z.object({
  to: z.enum(['archivist', 'regulatory', 'legal',
    'financial', 'synthesis']),
  intent: z.string(), // what + why
  authority: z.string(), // what may change
});
```

lib/orchestration/workflow.ts

```
// the agents decide who does what next
for (const d of delegations) {
  await delegate(dealId, roomId, d.from, d.to,
    d.intent, d.authority, async () =>
      run(d.to, { ...ctx, delegation: d }));
  // → real Band task: processing → processed
}
```

WHAT THIS ENABLES

Delegation and recruitment are values the agents produce, not control flow we wrote. A zoning conflict can route Legal → Financial, and a contamination flag can recruit Environmental — entirely from context.

Watch the committee work, in real time.

● Cedar Wharf · diligence room

8 agents · live

■ Archivist → Regulatory

Extracted property facts. Flagging two missing documents: ALTA survey, Phase I.

■ Legal Risk

Contract references a neighbour access easement — **conflicts** with the title record.

Contradiction · Archivist vs Legal — negotiating...

● Environmental recruited by Regulatory

Phase I warranted. Remediation P50 ≈ \$2.4M.

task Legal → Financial: re-underwrite · **processed** ✓

DEAL MEMO

YELLOW

Proceed with three conditions precedent: resolve the easement, complete Phase I, and re-price to a **9.1% IRR** floor.

LIVE EVENT STREAM

```
thought regulatory: assessing flood zone...
tool_call environmental.simulate(trials=1e4)
tool_result P50 2.4M · P90 6.1M
handoff legal → financial
financial.recalculated 14.2% → 9.1%
```

Python specialists that do real quantitative work.

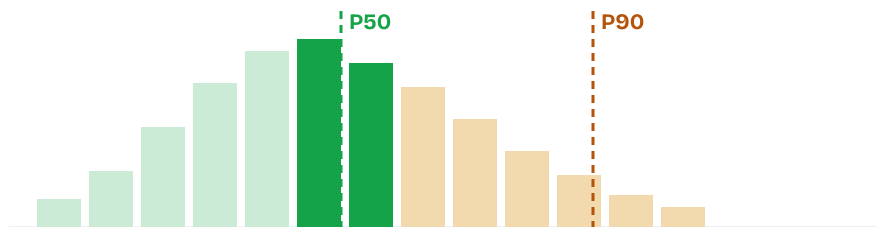
Three LangGraph state machines run seeded numpy Monte-Carlo — quantitative analysis the reasoning agents cannot fabricate. They are recruited live and post results back to the room.

CAPEX SPECIALIST — SAMPLE RUN · CEDAR WHARF

\$186M

P50 conversion cost

100% over budget at P50



ENVIRONMENTAL

Contamination-risk score and a remediation-cost distribution → a defensible **Phase I** recommendation.

INSURANCE / CATASTROPHE

Flood, wind and seismic event simulation → expected annual loss and a grounded premium estimate.

WHY PYTHON

The framework is not the point — the value is. numpy and LangGraph are the right tools for stochastic modelling, and Band makes them one committee regardless.

Numbers you can defend to an IC.

DETERMINISTIC UNDERWRITING

The IRR is not generated by a language model. The Financial agent's headline number is computed in code — DSCR and a levered 5-year IRR solved by **bisection** — so identical inputs always yield the same answer.

Baseline IRR

14.2%



After zoning cascade

9.1%

A critical upstream flag delegates a re-underwrite, and the IRR moves accordingly.

FULL AUDIT TRAIL

handoff Archivist → Regulatory: facts + 2 missing docs

conflict Legal vs Archivist easement — negotiated, resolved

recruit Regulatory pulled in Environmental specialist

task Legal → Financial: re-underwrite · processed ✓

memo Synthesis: **YELLOW** — 3 conditions precedent

Every line is persisted to Postgres and replayable — the answer to “why did we approve this?”

A large, underserved, high-value market.

TARGET USER

Institutional and mid-market commercial-real-estate acquirers — private-equity real-estate funds, REITs and family offices closing 20–100+ assets a year.

WHY NOW

Reliable multi-agent reasoning and a coordination layer like Band have only just become production-viable. Diligence is the ideal first wedge: structured, repetitive, and expensive to get wrong.

REVENUE MODEL

Enterprise SaaS seats plus per-deal pricing — on the order of a few thousand dollars per completed diligence, against the \$50k+ a single deal spends on outside counsel and consultants.

~\$24B

TAM

Annual US spend on diligence, legal and advisory across commercial-real-estate transactions.

~\$3.5B

SAM

Technology-forward institutional and mid-market acquirers reachable with a software product.

10×

Cost & time advantage

Minutes instead of weeks, at a fraction of per-deal advisory cost — with a defensible audit trail.

Market figures are internal estimates for illustration; the wedge is the per-deal value, not the headline TAM.

One room. A modern, cross-framework stack.



Coordinated by Band — the layer every agent reads from and writes to.

FRONTEND & APP



Next.js



React



TypeScript



Tailwind

PYTHON SPECIALISTS



Python



FastAPI



LangGraph



numpy

AI MODELS



Claude



Gemini



OpenAI

DATA & INFRA



Neon



Postgres



Drizzle



Vercel



An institutional diligence committee where Band is not plumbing — it is what makes eight agents a team.

Real coordination, real cross-framework value, and numbers an investment committee can defend.

...and nothing hardcoded.